Knowledge Graphs of Bank Volatility in the COVID-19 Crisis

Samson LAM

March 2026

Knowledge graphs

- A knowledge graph represents entities as vertices and their relationships as edges.
- Chen and Zhang (2024) assessed systemic risk in US banking with a sparse precision matrix over the leading financial firms, treating non-zero off-diagonal entries as edges.
- They compared the graph during the 2008 financial crisis with the graph outside it.
- For the liquidity discount index of Chen et al. (2016), they found firms more closely interconnected during the crisis.
- This supports the idea that a common macroeconomic shock displaces idiosyncratic risk.
- For volatility, they reported that the non-crisis period produced a more complex network than the crisis period.

Research question

Question

How does the knowledge graph of bank volatility inside the 2020 pandemic window compare with the graph outside it, for 70 US-listed banks at daily frequency from 2017 to 2021?

Finding

1. At a common cut-off $c = 0.035$, none of the 25 largest banks are isolated during the crisis, and the network holds a dense core.
2. This aligns with the expectation that firms tend to become more interconnected during crisis periods when common macroeconomic shocks overshadow idiosyncratic risks.
3. This is a claim about weak links only, and a large share of strong conditional dependence in this sample occurs in the non-crisis period.
4. At $c = 0.050$, there are 8 edges inside the crisis against 84 outside it, and the surviving structure matches the geographic split of the non-crisis graph.

Data

Period	Start	End	Observations
Full study	June 2018	December 2021	883
Crisis	December 2019	June 2020	124
Non-crisis	June 2018	November 2019	759
	June 2020	December 2021	

Definition of the subperiods.

- The universe is 70 bank stocks listed in the United States, from the market capitalization ranking of *Stock Analysis*.
- Daily adjusted closing prices run from May 2017 to December 2021.
- The crisis window follows the acute phase of the market shock identified by Liu et al. (2020), and the non-crisis window is the remainder of the study period.

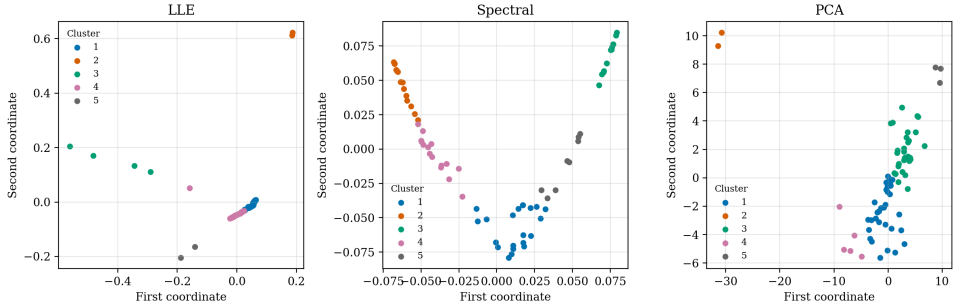
Embedding

$$z_{i,t} = s_i^{-1}(\sigma_{i,t} - \bar{\sigma}_i)$$

- $z_{i,t}$ is the standardized volatility within each subsample.
- $\sigma_{i,t}$ is the rolling sample volatility of bank i on day t, the trailing standard deviation over 1 year (252 trading days).
- $\bar{\sigma}_i$ and s_i are the mean and the standard deviation of $\sigma_{i,t}$ over the subsample.

- We map the resulting panel into two dimensions with locally linear embedding (LLE), spectral embedding and principal component analysis (PCA).
- The first two are eigenmaps preserving local geometry, and the third is the linear baseline.
- A k-means partition with $k = 5$ colors the vertices and serves as an independent check on the embedding, as in Chen and Zhang (2024).

Embeddings of the standardized volatility



Embeddings of the standardized volatility of all 70 banks over the study window, colored by a k-means partition with $k = 5$.

- In the LLE, most banks cluster in a dense mass around (0.0, 0.0), with only a few outliers.
- Spectral embedding reveals a curved manifold that suggests nonlinear relationships among the banks' volatility patterns, and PCA is the most dispersed of the three.

Graphical LASSO

$$\hat{\Theta} = \arg\min_{\Theta \succ 0} \{\mathrm{tr}(S\Theta) - \log\det\Theta + \alpha\|\Theta\|_1\}$$

- We identify conditional independence with the graphical least absolute shrinkage and selection operator (LASSO), which estimates the precision matrix as $\hat{\Theta}$.
- S is the sample correlation matrix of $\{z_{i,t}\}$ over a subsample, and the penalty α is chosen by three-fold cross-validation.
- A non-zero off-diagonal entry means that two banks remain dependent once every other bank in the cross section is conditioned on.

Partial correlation

$$\hat{\rho}_{ij} = -\frac{\hat{\Theta}_{ij}}{\sqrt{\hat{\Theta}_{ii}\hat{\Theta}_{jj}}} \in [-1, 1], \qquad E(c) = \#\left\{(i,j) : i < j,\ |\hat{\rho}_{ij}| > c\right\}$$

- The entries of $\hat{\Theta}$ are not comparable across two separate fits, since each carries the scale of its own diagonal.
- The edge weight is therefore the partial correlation $\hat{\rho}_{ij}$, and $E(c)$ is the number of edges at a cut-off c.
- The 25 largest banks are carried into the spring layout, where edge width is proportional to $|\hat{\rho}_{ij}|$ and vertex color is the cluster label.

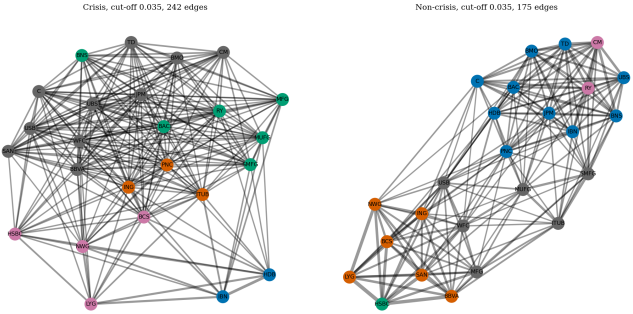
Fitted precision matrices

	Crisis	Non-crisis
Selected penalty α	0.600	0.599
Median $\lvert\hat{\rho}_{ij}\rvert$	0.0397	0.0386
Standard deviation of $\lvert\hat{\rho}_{ij}\rvert$	0.0061	0.0199

Fitted precision matrices of the 25 largest banks.

- The two fits are penalized to the same degree.
- The median $\lvert\hat{\rho}_{ij}\rvert$ is nearly the same in the two subsamples, but its standard deviation is three times as large outside the crisis.

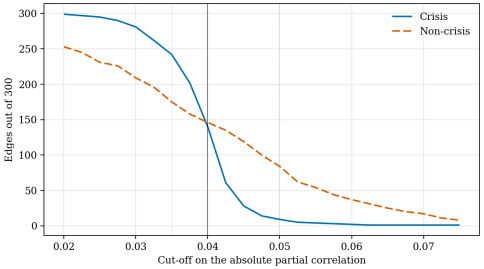
Knowledge graphs of the 25 largest banks



Knowledge graphs of the 25 largest banks at the common cut-off $c = 0.035$.

- No bank is isolated during the crisis, and the network holds a dense core.
- The non-crisis graph is visually sparser and more organized, separating into a North American group and a British and continental European group.

Edges against a common cut-off



Edges against a common cut-off on the absolute partial correlation. The solid vertical line marks the crossing at $c = 0.040$ and the dotted lines mark $c = 0.035$ and $c = 0.050$.

- Below $c = 0.040$ the crisis graph is denser, and above it the ordering reverses and widens.
- There are 8 crisis edges against 84 at $c = 0.050$, and 1 against 8 at $c = 0.075$.
- The claim that banks are more connected during the crisis is therefore a claim about weak links only.

Conclusion

- We transfer the framework of Chen and Zhang (2024) to the pandemic episode, comparing knowledge graphs of bank volatility inside and outside the crisis window.
- At $c = 0.035$, none of the 25 largest banks are isolated during the crisis, and the network holds a dense core.
- This aligns with the expectation that firms tend to become more interconnected during crisis periods when common macroeconomic shocks overshadow idiosyncratic risks.
- This is a claim about weak links only. At $c = 0.050$, there are 8 edges inside the crisis against 84 outside it.
- The surviving structure matches the geographic split of the non-crisis graph, consistent with the volatility result of Chen and Zhang (2024).

References

- Chen, R.-R., Lin, W. and Wei, H. (2016). A Liquidity Index. *Journal of Fixed Income*, 25(4), 5–19.
- Chen, R.-R. and Zhang, X. (2024). From Liquidity Risk to Systemic Risk: A Use of Knowledge Graph. *Journal of Financial Stability*, 70, 101195.
- Liu, H., Manzoor, A., Wang, C., Zhang, L. and Manzoor, Z. (2020). The COVID-19 Outbreak and Affected Countries Stock Markets Response. *International Journal of Environmental Research and Public Health*, 17(8), 2800.

Code and replication data are available at *GitHub*.